Summary

Of 16 planned features, 11 are completed (two of them with changes), 2 were changed after testing, 1 is partial and 2 are not implemented. Every row below is checked against the approved proposal and the code in the repository. Paths are relative to the repo root.

Feature table

Planned feature	Intended purpose	Current status	Evidence
Dataset collection, cleaning and preprocessing	Build a clean training set from 332 patients	Completed	Notebook sections 3–4: 3,359 photos reduced to 3,310 after removing duplicates and unclear labels; timestamp mask; resize to 256×192; patient-grouped 5-fold split
Binary Normal/Abnormal model	Classify colposcopy images	Completed, modest accuracy	Notebook sections 6–12; backend/app/services/predictor.py; patient-level AUC 0.625 (95% CI 0.565–0.688)
Transfer learning with fine-tuning	Cope with a small dataset	Tested, not used in the final model	Notebook sections 8–10: four pretrained backbones scored 0.552–0.576, below the 0.625 baseline
Before- and after-acid images	Use both photo types	Changed: after-acid only	Notebook introduction: after-acid photos carried the signal
Grad-CAM heatmaps	Show which regions drove the decision	Completed	predictor.py (_gradcam, _overlay); GET /images/{id}/heatmap; frontend/src/pages/Result.jsx

Certainty metrics	Give a confidence score and level per result	Completed	predictor.py (confidence_score, certainty_level); result page
Plain-language explanation	Readable text for health workers	Completed	predictor.py (_explanation); "What this means" panel on the result page
Web upload and results page	Lightweight platform for clinicians	Completed	Dashboard.jsx, Result.jsx, backend/app/routers/images.py
Sign up, log in, JWT sessions	Secure clinician access	Completed	routers/auth.py, security.py; 35 automated auth tests
Dashboard and history	Revisit past results	Completed	History.jsx, GET /images
Image quality check	Reject poor images before analysis	Completed	services/quality.py; size, type and pixel limits; 17 input tests
Database and file storage	Store clinicians, images, results and heatmaps	Completed, with changes	models.py, services/storage.py; integer IDs instead of UUIDs

Evaluation with measurable metrics	Judge model performance	Partial	Notebook sections 6–11: AUC, sensitivity 0.67 and specificity 0.53 at a provisional threshold. "Reduced reliance on specialists" needs a clinical study, which the scope excludes
Severity grading beyond binary	Finer, Bethesda-style output	Not implemented	Scope section 1.5 fixes the task as binary
Testing and debugging	Verify the whole system	Completed	85 backend tests; 24 browser tests; load test with 0 failures at 5, 20 and 40 users
Hosted deployment	Run beyond one laptop	Not implemented	Clinical deployment is out of scope; the app runs locally

Scope changes and why

- **After-acid photos only.** The earlier runs showed that after-acid photos carried the signal.
- **Small CNN instead of transfer learning.** It scored higher in cross-validation (AUC 0.625 against at most 0.576).
- **Integer IDs.** They are simpler with SQLite, and every image route checks ownership.
- **No patient identifiers or original file names stored.** This is a privacy choice, and the scope excludes real patients.
- **Image resolution and qualityPassed fields not saved.** They appear in the class diagram but are not kept in the database.
- **Model runs before the database write.** A load test showed "database is locked" errors at 20 users, so the sequence diagram changed. After the fix, 0 of 60 uploads failed.

Known limitations and future work

- The model is a research prototype with modest accuracy. It is not validated for clinical use.

- Sign-up checks that an email is correctly formatted. It does not verify that the mailbox exists, and there is no confirmation email or password reset.
- Only 640×480 after-acid photos are accepted.
- Speed and memory figures came from a 2-core cloud machine with stand-in models. They must be rerun on the project laptop with the real models.
- Browser tests ran on Chromium and a Pixel 5 emulation. Firefox and Safari (WebKit) are still to be run.
- Future work: email verification, password reset, login throttling, severity grading and a clinical validation study.